

Tax Evasion and Productivity

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Recap

Last meeting: What is an interesting question to wrap up the JMP?

- ▶ Counterfactual question: Optimal tax rate

Today:

- ▶ SOC of firm's optimization problem
- ▶ Functional forms for detection probability and tax evasion cost
- ▶ Estimation strategy

Introduction

- ▶ What's the question?
- ▶ Why should you care?

Contribution

- ▶ What do I do?
- ▶ How do I do it?

Overview of Results

- ▶ What do I find?
- ▶ Why is it important?

Colombia

Model

The Model

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & (1 - q(e_{it})) \left[P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it} - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \tau (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t (M_{it} + e_{it})) \right] \\ & + q(e_{it}) \left[(1 - \tau) (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \phi \rho_t \tau e_{it} \right] - \kappa_{it}(e_{it}, \omega_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^\beta \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{1}$$

Important Assumptions

- ▶ Following seminal work of Allingham and Sandmo (1972), tax authority audits firms at random and always detects tax evasion
 - ▶ The probability of detection, $q(e_{it})$, is a function of the level of tax evasion, e_{it} . It might depend on other exogenous variables such as the monitoring technology of the tax authority, for example.
 - ▶ Impose assumption from Empirical model which assumes that choosing M does not enter the probability of detection or the cost of evasion. In other words, choosing M is independent of choosing e.

Important Assumptions (Cont.)

The cost of evasion, $\kappa_{it}(e_{it}, \omega_{it})$: any resources of the firm allocated to the tax evasion activity that could have been employed in a productive activity (e.g. manager's time spent on tax evasion instead of managing the firm).

- ▶ Unlike the penalty of tax evasion, ϕ , it does not depend on whether you are detected or not
- ▶ Increasing in the level of tax evasion, e_{it}
- ▶ Decreasing in firm's productivity, ω_{it} , but spikes at the top of the productivity distribution.

Important Assumptions - discussion on κ

- ▶ More productive firms have more resources to allocate to tax evasion, but since huge firms have access to more sophisticated legal tax avoidance strategies, the cost of illegal tax evasion is higher for the most productive firms.
- ▶ It could also be the case that the probability of detection is flat with respect to productivity except for huge (top productive) firms which could have a probability close to 1.
- ▶ Empirically, without data on cost of evasion or auditing information, q and κ are not separately identified. During estimation, we assume functional forms that allow us to reproduce the empirical patterns of tax evasion across the productivity distribution.

Model (Cont.)

The firms' problem (Equation 1) can be rewritten as follows,

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & \mathbb{E}[\pi_{it} | \Gamma] = (1 - \tau) (P_t \mathbb{E}[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \\ & + [1 - q(e_{it})] (\rho_t \tau e_{it}) - q(e_{it}) \phi \rho_t \tau e_{it} - \kappa_{it}(e_{it}, \omega_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{2}$$

First Order Condition (FOC)

An interior solution leads to following first-order conditions,

$$\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{-(1-\beta)} \exp(\omega_{it}) = \frac{\rho_t}{P_t} \quad (3)$$

$$\tau \rho_t (1 - (1 + \phi) (q(e_{it}) + q'(e_{it})e_{it})) = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}} \quad (4)$$

(ε_{ijt} is measurement error, so $\mathcal{E} \equiv E[\exp(\varepsilon_{ijt})] = 1$)

Tax Evasion

To better understand the incentives to overreport inputs, we can rewrite Equation 4 in terms of the evasion elasticity of the probability of detection, $\xi_{q,e} = \frac{\partial q(e_{it})}{\partial e_{it}} \frac{e_{it}}{q(e_{it})}$, as follows,

$$\tau \rho_t \left(1 - (1 + \phi) (1 + \xi_{q,e}) q(e_{it}) \right) = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}} \quad (5)$$

Corner Solution

Note that some firms will optimally choose not to evade, $e_{it} = 0$.

For these firms, the marginal cost of tax evasion is greater than the marginal benefit for any non-zero level of tax evasion.

This implies that for these firms, Equation 4 becomes,

$$\tau \rho_t \leq \frac{\partial \kappa_{it}(0, \omega_{it})}{\partial e_{it}} \quad (6)$$

Second Order Condition (SOC)

To guarantee a unique solution, the hessian matrix of the firm's problem must be negative definite.

$$\nabla^2 \mathbb{E}[\pi_{it} | \Gamma] = \begin{bmatrix} \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial e_{it}^2} & 0 & 0 & 0 \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it}^2} \end{bmatrix} \quad (7)$$

- ▶ For a negative definite hessian matrix, the leading principal minors must alternate signs,
 - ▶ odd order minors must be negative and even-order minors positive.
- ▶ In our case, since
 - ▶ 1) the production function is strictly concave (CD) and
 - ▶ 2) choosing the level of tax evasion does not depend on the production function inputs; second cross derivatives are zero,
 - ▶ we would only have to check the first leading principal minor —the second derivative with respect to e_{it} — is negative,

$$\frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial e_{it}^2} = -\tau \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) - \frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} < 0 \quad (8)$$

For strict concavity, we need only one of the following two conditions to be strictly positive,

$$\frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} \geq 0 \quad (9)$$

$$\tau \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) \geq 0 \quad (10)$$

The second condition can be simplified to

$$-\frac{q''(e_{it})}{q'(e_{it})}e_{it} < 2 \quad (11)$$

since economic theory implies that $q'(e_{it}) > 0$ and $e_{it} > 0$, this condition imposes a **restriction on the curvature** of the detection function, $q(e_{it})$

- ▶ **Linear** detection **always** satisfies the SOC, since $q''(e_{it}) = 0$.
- ▶ **Convex** also satisfies the SOC **always**, since $q''(e_{it}) > 0$.
- ▶ **Concave** works if not too concave, i.e. **inequality holds**.

What does this inequality imply in practice?

With $\phi = 0$, logs of the interior FOC $\tau\rho_t(1 - q - q'e) = \partial\kappa/\partial e$ define

- ▶ the **log marginal benefit** $h(e) \equiv \ln(\tau\rho_t) + \ln(1 - q(e) - q'(e)e)$,
- ▶ with FOC $h(e) = \ln(\partial\kappa/\partial e)$.

	$q(e)$	SOC	$h(e) - \ln(\tau\rho_t)$	FOC ceiling
Linear	$\min\{\lambda e, 1\}$	$0 < 2$	$\ln(1 - 2\lambda e)$	$\lambda e < \frac{1}{2}$
Concave	$1 - e^{-\lambda e}$	$\lambda e < 2$	$-\lambda e + \ln(1 - \lambda e)$	$\lambda e < 1$
Convex	$\min\{(\lambda e)^k, 1\}$ $k > 1$	$1 - k < 2$	$\ln(1 - (1+k)(\lambda e)^k)$	$\lambda e < (1+k)^{-1/k}$

- ▶ All satisfy the SOC on their interior (exp's $\lambda e < 2 \iff \lambda e < 1$);
- ▶ All impose a restriction on λe .

A detection technology: the audit threshold

- ▶ The firm chooses overreporting e .
- ▶ The tax authority draws a threshold $T \sim U[0, 1/\lambda]$ and audits the firm if $e \geq T$.
 - ▶ An audit always uncovers evasion (Allingham and Sandmo 1972).
 - ▶ T is random to the firm (audit selection, third-party cross-checks, which inspector).

$$q(e) = \Pr(T \leq e) = \min\{\lambda e, 1\}.$$

- ▶ **Threshold scale:** $1/\lambda$ is the largest overreporting any audit draw tolerates; the *average* audit-triggering threshold is $1/2\lambda$.
- ▶ **Evasion bound:** the FOC stops firms at $\lambda e < \frac{1}{2}$
 - ▶ the most aggressive evader overreports below the *average* audit-triggering threshold,
 - ▶ at threshold, $q(1/2\lambda) = 50\%$ (a coin flip)

If $q(e)$ is concave, $T \sim \text{Exp}(\lambda)$ and $q(e) = 1 - e^{-\lambda e}$, mean threshold $1/\lambda$, detection $\approx 63\%$ at $e = 1/\lambda$.

How likely is the most aggressive evader caught?

At the FOC ceiling, the largest evasion drives the marginal benefit to zero,
 $q(b) + q'(b)b = 1$.

	b	$q(b)$	$\xi_{q,e} _{e=b}$	$E[T]$
Linear	$\frac{1}{2\lambda}$	50%	1 (unit elastic)	$\frac{1}{2\lambda}$
Exponential	$\frac{1}{\lambda}$	63%	$\frac{\lambda b}{e^{\lambda b} - 1} < 1$ (inelastic)	$\frac{1}{\lambda}$
Power	$\frac{1}{\lambda(1+k)^{1/k}}$	$\frac{1}{1+k} < 50\%$	$k > 1$ (elastic)	$\frac{k}{\lambda(k+1)}$

- ▶ $\xi_{q,e} = q'(e)e/q(e)$ is the evasion elasticity of the detection probability
- ▶ $\frac{1}{e-1} = 0.58$

Discussion

- ▶ Power (convex) form produces the lowest average audit-triggering threshold,
 - ▶ Exponential (concave) the largest.
 - ▶ Linear is intuitive: at max evasion, the tax authority flips a coin to audit/caught the firm
- ▶ An increase in 1% in the level of evasion increases the probability of being caught, by
 - ▶ 1% if linear (constant)
 - ▶ more than 1% if convex (constant)
 - ▶ less than 1% if concave (increasing), $\xi_{q,e}|_{e=0} = 0$ and $\xi_{q,e}|_{e=1/\lambda} = 0.58$.
- ▶ The extra dollar the firm hides increase its probability of detection more than the previous dollar.
- ▶ For linear and exponential, the FOC bounds evasion *at* the expected value of T
 - ▶ For power, FOC bounds evasion *below* its expected value: At $k = 2$, $\frac{1}{\lambda\sqrt{3}} < \frac{2}{\lambda 3}$

Estimation and recommendation

- ▶ **Linear** and **exponential** carry a single parameter λ
- ▶ **Power** $(\lambda e)^k$ adds a parameter, curvature k

Do **linear** and **power**,

- ▶ Start with **linear** ($k = 1$). Linear is the $k = 1$ case of the power form (ceiling $(1+k)^{-1/k} = \frac{1}{2}$).
- ▶ Then, estimate power (λ, k) , imposing feasibility $\lambda e < (1+k)^{-1/k}$ for *every* firm. Test if $k = 1$.

Empirical Strategies

- ▶ Testing for the presence of tax evasion by industry
 - ▶ Robust to different specifications
- ▶ Regression
 - ▶ Useful to study tax evasion heterogeneity across firms and over time
- ▶ Deconvolution
 - ▶ Recover distribution of tax evasion and productivity
 - ▶ Estimation of unbiased Production Function

Data

Testing

Empirical Strategy

Functional Forms

The the probability of detection, $q(e_{it})$, and the cost of evasion, $\kappa_{it}(e_{it}, \omega_{it})$, functions should display the following properties to be consistent with the observed patterns in the data and with theory:

- ▶ $q(e_{it}) = \min\{\lambda e, 1\}$
 - ▶ $q(e_{it}) \in [0, 1]$, $q(0) = 0$, $q(\infty) = 1$, and $q(e_{it})$ is increasing in e . Since I do not have variance in govt effort, a will be constant $q(e_{it})$
- ▶ $\kappa_{it}(e_{it}, \omega_{it}) = e_{it} \mathbf{e}^{\delta_0 - \delta_1 \omega_{it} + \delta_2 \omega_{it}^2 + \psi_{it}}$
 - ▶ $\kappa_{it}(e_{it}, \omega_{it}) \in [0, \infty)$, $\kappa_{it}(0, \omega_{it}) = 0$, $\frac{\partial \kappa_{it}(0, \omega_{it})}{\partial e_{it}} \geq 0$.
 - ▶ $\frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial \omega_{it}} \leq 0$ and $\frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial \omega_{it}^2} \geq 0$ to be consistent with the observed patterns in the data.
 - ▶ U-shape not ideal, but simple and captures the idea of decreasing cost at high ω
 - ▶ Corporations out of the FOC
 - ▶ ψ_{it} idiosyncratic shock to the cost of evasion to explain tax evasion heterogeneity observed in empirical analysis

FOC

Using the functional forms defined above, the FOCs wrt tax evasion become for unincorporated firms,

$$\tau\rho_t(1 - 2\lambda e_{it}) = \mathbf{e}^{\delta_0 - \delta_1\omega_{it} + \delta_2\omega_{it}^2 + \psi_{it}}$$

Taking logs and defining

$$h(e_{it}) \equiv \ln(\tau\rho_t) + \ln(1 - \lambda 2e_{it}), \quad e_{it} \in (0, \frac{1}{2\lambda}),$$

the evasion FOC become

$$\psi_{it} = h(e_{it}) - \delta_0 + \delta_1\omega_{it} - \delta_2\omega_{it}^2 \tag{12}$$

How to estimate?

I do not observe e_{it} nor ω_{it} at the firm level.

- ▶ However, using the estimated parameters of the production function, we can rewrite the FOC in terms of data, parameters, and *only one* independent unobservable ε , as follows,

Use the estimated parameters of the production function to form,

- ▶ $\tilde{\mathcal{V}}_{it} \equiv \ln \left(\frac{\rho_t(M_{it} + e_{it})}{P_t Y_{it}} \right) - \ln \beta = \ln \left(\frac{M_{it}^*}{M_{it}^* - e_{it}} \right) - \varepsilon_{it}$
- ▶ $\tilde{\mathcal{W}}_{it} \equiv y_{it} - \beta(m_{it}^* - \mathcal{V}_{it}) - \alpha_K k_{it} - \alpha_L l_{it} = \omega_{it} + (1 - \beta)\varepsilon_{it}$
- ▶ $\tau \rho_t$ is *computed from data*

Recast in terms of one unobservable

For each value of ε everything else is pinned down:

$$e(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})$$

$$\omega(\varepsilon) = \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon$$

$$\psi(\varepsilon) = h(e(\varepsilon)) - \delta_0 + \delta_1 \omega(\varepsilon) - \delta_2 \omega(\varepsilon)^2.$$

- Estimate by Maximum Simulated Likelihood (MSL) or Method of Simulated Moments (MSM)

- ▶ Sample from $\varepsilon^{(s)} \sim \hat{f}_\varepsilon$
- ▶ Form $\psi(\varepsilon^{(s)}) = h(e(\varepsilon^{(s)})) - \delta_0 + \delta_1 \omega(\varepsilon^{(s)}) - \delta_2 \omega(\varepsilon^{(s)})^2$
- ▶ Find instruments such that

$$E[\psi(\varepsilon^{(s)})|Z] = 0$$

$$\begin{aligned}f_{e,\omega}(e, \omega) &= \int f_{e,\omega,\varepsilon}(e, \omega, \varepsilon) d\varepsilon \\&= \int f_{e|\omega,\varepsilon}(e|\omega, \varepsilon) f_{\omega|\varepsilon}(\omega|\varepsilon) f_{\varepsilon}(\varepsilon) d\varepsilon \\&= \int f_{e|\omega}(e|\omega) f_{\omega}(\omega) f_{\varepsilon}(\varepsilon) d\varepsilon \\&= \int f_{\psi}(\psi(\varepsilon)) f_{\omega}(\omega(\varepsilon)) f_{\varepsilon}(\varepsilon) d\varepsilon\end{aligned}$$

The object to estimate is f_{ψ} and $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$, since I know f_{ω} and f_{ε}

Simulated log-likelihood

$$\mathcal{L}(\theta, f_\psi) = \sum_{i \in \text{Unincorp}} \ln \left(\int f_\psi(\psi(\varepsilon)) f_\omega(\omega(\varepsilon)) f_\varepsilon(\varepsilon) d\varepsilon \right) \quad (13)$$

Alternatively, cast moment conditions in terms of M

We can recast the moment conditions in terms of one unobservable M

$$\varepsilon_{it} = \ln(\beta) - \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right)$$

$$e_{it} = M_{it}^* - M_{it}$$

$$\psi_{it} - \delta_1 \varepsilon_{it} = h(M_{it}^* - M_{it}) - \delta_0 + \delta_1 \Omega_{it}$$

- ▶ where $\Omega_{it} \equiv y_{it} - \beta \ln M_{it} - \alpha_K k_{it} - \alpha_L l_{it}$
- ▶ Either deconvolve M from $\mathcal{V}_{it}$ using f_ε and recover $f_{M|M^*}$,
 $\int f_\varepsilon (\ln (M_{it}^*/M_{it}) - \mathcal{V}_{it}) f_{M|M^*} (M_{it}^* - M_{it}) dM$ and simulate $M_{it}^{(s)} \sim \hat{f}_{M|M^*}$
- ▶ or simulate $M_{it}^{(s)} = M_{it}^* \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon^{(s)}} \sim \hat{f}_\varepsilon$

We can find instruments and use Method of Simulated Moments (MSM)

$$\mathbb{E}[\varepsilon_{it}|Z_1] = 0$$

$$\mathbb{E}[e_{it}|\text{Corp}] = 0$$

$$\mathbb{E}[\mathbb{1}\{e_{it} > 0\} - 1|\text{Unincorp}] = 0$$

$$\mathbb{E}[\psi_{it} - \delta_1 \varepsilon_{it}|Z_2] = 0$$

A note on the empirical strategy

- ▶ Empirical strategy of first part should be consistent with the updated theoretical model
 - ▶ In the theoretical model, I'm using $M_{it}^* = M_{it} + e_{it}$
 - ▶ In current empirical strategy, I'm using $M_{it}^* = M_{it} \exp\{\tilde{e}_{it}\}$
- ▶ Why? Because I will use the estimated parameters and distributions of the first stage (production function parameters and the distributions of the error term, tax evasion, and productivity)
- ▶ Since, it is easier to adjust the estimation strategy of the first stage to be consistent with the theoretical model, I will stick with the theoretical model definition, and adjust the estimation strategy of the first stage accordingly.

Adjusting the first stage estimation

- I can still recover the output elasticity of materials, β , and the distribution of the error term, ε_{it} , using Corporations, the truth-reporters. Since for them, $M_{it}^* = M_{it}$, and thus,

$$\ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) = \ln(\beta) - \varepsilon_{it}$$

Adjusting the first stage estimation

► For evaders, this implies that,

$$\begin{aligned}\mathcal{V}_{it} &\equiv \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) - \ln(\beta) \\ &= \ln \left(\frac{\rho_t (M_{it} + e_{it})}{P_t Y_{it}} \right) - \ln(\beta) \\ &= \ln \left(\frac{\rho_t (M_{it} + e_{it})}{P_t Y_{it}} \right) - \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) + \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) - \ln(\beta) \\ &= \ln \left(\frac{M_{it} + e_{it}}{M_{it}} \right) - \varepsilon_{it}\end{aligned}\tag{14}$$

Adjusting the first stage estimation

- ▶ Rewriting Equation 14 in terms of M_{it}^* , I can use it to recover the distribution of tax evasion, $f_e(e)$, by deconvolving the distribution of ε_{it} from the distribution of $\mathcal{V}_{it}$.

$$\mathcal{V}_{it} = \ln \left(\frac{M_{it}^*}{M_{it}^* - e_{it}} \right) - \varepsilon_{it} \quad (15)$$

- ▶ Note the term $\ln \left(\frac{M_{it} + e_{it}}{M_{it}} \right)$ is equivalent to the previous definition of $\tilde{e}_{it}$,

$$M_{it} \exp\{\tilde{e}_{it}\} = M_{it} + e_{it} \implies \tilde{e}_{it} = \ln \left(\frac{M_{it} + e_{it}}{M_{it}} \right)$$

Production Function Estimation

- To recover the remaining production function parameters, the original strategy is still valid. To see why, note that,

$$\begin{aligned}\mathcal{W}_{it} &\equiv y_{it} - \beta(m_{it}^* - \mathcal{V}_{it}) \\ &= y_{it} - \beta(m_{it}^* - \ln\left(\frac{M_{it} + e_{it}}{M_{it}}\right) + \varepsilon_{it}) \\ &= y_{it} - \beta(m_{it}^* - \ln(M_{it}^*) + \ln(M_{it})) - \beta\varepsilon_{it} \\ &= y_{it} - \beta m_{it} - \beta\varepsilon_{it} \\ &= \alpha_K k_{it} + \alpha_L l_{it} + \omega_{it} + (1 - \beta)\varepsilon_{it}\end{aligned}$$

Therefore, the previous approach is still valid to recover the remaining production function parameters and the distribution of the productivity shock $(\alpha_K, \alpha_L, \omega_{it})$.

Implicit Assumptions

- ▶ The functional forms for $q(e)$ and $\kappa(e)$ are correctly specified and do not change over time
- ▶ No entry, no exit
- ▶ Corporations never evade
- ▶ Partial equilibrium, Inelastic demand, (No GE effects)
 - ▶ Tax increases do not change optimal input choices and output. FOC of materials does not depend on tax rate
 - ▶ Firms are already assumed to be price takers

What about τ ?

- ▶ The data includes sales taxes paid on purchases, $\tau \rho_t M_{it}^*$ and sales taxes paid on sales, $\tau P_t Y_{it}$.
- ▶ I can estimate at the firm level the sales tax rate on sales $\tau_{S,it} = \frac{\tau P_t Y_{it}}{P_t Y_{it}}$ and the sales tax rate on purchases $\tau_{P,it} = \frac{\tau \rho_t M_{it}^*}{\rho_t M_{it}^*}$.
- ▶ I can also estimate the reported effective sales tax rate $\tau_{R,it} = \frac{\tau P_t Y_{it} - \tau \rho_t M_{it}^*}{P_t Y_{it}} = \tau_{S,it} - \tau_{P,it} \frac{\rho_t M_{it}^*}{P_t Y_{it}}$
- ▶ and the *corrected* effective sales tax rate $\tau_{C,it} = \tau_{S,it} - \tau_{P,it} \frac{\rho_t M_{it}^*}{P_t Y_{it}} = \tau_{S,it} - \tau_{P,it} \beta e^{-\varepsilon_{it}}$
 - ▶ I use in regressions $\tau_{\tilde{C},it} = \tau_{S,it} - \tau_{P,it} \beta$

Is the variation in the different estimated tax rates relevant

- ▶ I use the following regression model

$$\mathcal{V}_{it} = \gamma_j + \gamma_t + \tau_{X,it} + v_{it}$$

- ▶ where γ_j are industry fixed effects,
- ▶ γ_t are year fixed effects, and
- ▶ $\tau_{X,it}$ is one of the estimated tax rates ($\tau_{S,it}$, $\tau_{P,it}$, $\tau_{R,it}$, or $\tau_{\tilde{C},it}$).

Variation in estimated τ predicts variation in tax evasion across firms

Dependent Variable: Model:	(1)	(2)	$\mathcal{V}$ (3)	(4)	(5)
<i>Variables</i>					
τ_S	-0.2804* (0.1333)		0.5053*** (0.1422)		
τ_P		-1.890*** (0.1252)	-2.026*** (0.1258)		
τ_R				-1.387* (0.6846)	
τ_C					0.9046*** (0.1803)
<i>Fixed-effects</i>					
Industry (3-digit)	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	40,490	40,505	40,490	40,490	40,490
R ²	0.04767	0.09016	0.09215	0.07638	0.05335
Within R ²	0.00068	0.04532	0.04736	0.03081	0.00665

Clustered (plant & Year) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Couterfactual Question

What Would Be The Optimal Tax Rate?

The government faces a trade-off:

- ▶ as the government increases the tax rate, the tax revenue increases
- ▶ as the tax rate increases, the marginal benefit of evading taxes increase leading to higher levels of tax evasion which reduces tax revenue

What is the optimal rate that maximizes expected tax revenue for the government?

What is the tax rate that lead to median (media) tax evasion of zero?

The Government's Problem

The government wants to choose the sales tax rate τ that maximizes the expected tax revenue from the sum of all firm i

$$\max_{\tau} \sum_i q(e)\tau [P_t Y_{ijt} - \rho_t M_{ijt}] + (1 - q(e))\tau [P_t Y_{ijt} - \rho_t (M_{ijt} + e_{ijt})]$$

Simplifying,

$$\max_{\tau} \sum_i \tau [P_t Y_{ijt} - \rho_t M_{ijt} - (1 - q(e))\rho_t e_{ijt}]$$

where $q(e)$ is the probability of detection, which is a function of the evasion level, e , and the government effort a .

for firm i ,

$$\begin{aligned} & [P_t Y_{ijt} - \rho_t M_{ijt} - (1 - q(e)) \rho_t e_{ijt}] \\ & + \tau \left[-(1 - q(e)) \rho_t \frac{\partial e_{ijt}}{\partial \tau} + \rho_t e_{ijt} \frac{\partial q(e)}{\partial e} \frac{\partial e_{ijt}}{\partial \tau} \right] = 0 \end{aligned}$$

simplifying,

$$\begin{aligned} P_t Y_{ijt} - \rho_t M_{ijt} &= (1 - q(e)) \rho_t e_{ijt} \\ &+ \tau \rho_t \frac{\partial e_{ijt}}{\partial \tau} \left[(1 - q(e)) - \frac{\partial q(e)}{\partial e} e_{ijt} \right] \end{aligned}$$

Numerical Strategy

1. Define functional forms for the probability of detection, $q(e)$, and the cost of evasion, $\kappa(e)$, that are consistent with the observed patterns in the data and with theory
2. Estimate the parameters of these functions using moments from the FOC of the firm's optimization problem (Equation 4) and the deconvolved distribution of tax evasion

3. Given the estimated parameters and a proposed tax rate, find the simulated distribution of tax evasion $f_{\tilde{e}}$ such that
 - i. the FOC wrt e of the firm's optimization problem (Equation 4) holds on average with the new $\tilde{\tau}$ and
 - ii. the moments derived from $E[\varepsilon|\tilde{e}] = 0$ also hold (e.g. $E[\varepsilon\tilde{e}] = 0$, $E[\varepsilon\tilde{e}^2] = 0$, etc.)
4. Compute the expected tax revenue for the proposed tax rate using the simulated distribution of tax evasion
 - i. Note that since I have $f_e(e)$, I also have the distribution of the true inputs since $M = M^* - e$, and thus I can compute the expected tax revenue for a simulated tax evasion $\tilde{e}$ corresponding to the proposed $\tilde{\tau}$

$$E[R(\tilde{\tau})] = \sum_i \left[\tilde{\tau} \int [P_t Y_{ijt} - \rho_t(M_{ijt}^* - e)] f_e(e) de - \tilde{\tau} \int (1 - q(\tilde{e}, a)) \rho_t \tilde{e} f_{\tilde{e}}(\tilde{e}) d\tilde{e} \right]$$



5. Find the tax rate that maximizes expected tax revenue

The Geography of Tax Evasion

How Does a Firm's Location Affect its Tax Evasion?

Firms located in different regions might face different probabilities of detection and different tax rates, and thus different incentives to evade taxes.

- ▶ Tax rates:
 - ▶ Departments and municipalities in Colombia can set their own tax rates similar to sales tax, for example the Industry and Commerce Tax
 - ▶ Special tax regimes for firms located in certain regions, such as the Free Trade Zones (Zonas Francas) and the Special Economic Zones (Zonas Económicas Especiales)
 - ▶ Special ownership model in Atlantic coast
 - ▶ Some variation in tax rates is observed in the data

How Does a Firm's Location Affect its Tax Evasion?

- ▶ Probability of detection: Two hypothesis
 1. Firms located near the tax authority's offices might have access to the tax authority (lobbying, bribery, etc.) and thus face a lower probability of detection
 2. Firms located farther away from the tax authority's offices might face a lower probability of detection, because the auditing cost for the tax authority increases with distance, and thus they are less likely to be audited

Data

In the dataset from Colombia, I have two geographic variables:

- ▶ `section_country_code`: Section of the country (eg. Antioquia, Atlantico, Bogota D.E., etc.)
- ▶ `metro_area_code`: Metropolitan Area (eg. Bogota, Cali, Medellin, Barranquilla).

Metro Area Summary Statistics

Metro Area	N	Market Share	Avg. Sales Tax Rate	Med. Sales Tax Rate	Avg. Tax Evasion	Med. Tax Evasion
Bogotá D.C., Soacha	14345	35.3	7.4	8.9	4.2	10.3
Medellín, Valle de Aburrá	8147	19.9	7.6	9.2	4.6	10.7
Rest of Colombia	5895	14.9	4.6	2.0	0.7	10.2
Cali, Yumbo	4538	11.4	7.2	8.4	4.2	10.1
Barranquilla, Soledad	2896	7.3	5.6	5.9	2.5	13.3
Bucaramanga, Giron, Floridablanca	2311	5.3	5.7	6.0	5.9	13.8
Pereira, Santa Rosa de Cabal, Dosquebradas	1128	2.7	7.6	9.1	11.1	21.1
Manizales, Villamaria	639	1.6	7.0	8.1	3.2	7.2
Cartagena	606	1.6	4.6	2.7	11.6	20.0

Section of the Country Summary Statistics

Depto	N	Market Share	Avg. Sales Tax Rate	Med. Sales Tax Rate	Avg. Tax Evasion	Med. Tax Evasion
Bogotá D.C.	13954	34.3	7.3	8.9	4.3	10.3
Antioquia	8467	20.7	7.5	9.1	3.4	9.7
Valle del Cauca	5780	14.6	6.7	7.0	4.1	10.3
Atlántico	2922	7.3	5.6	5.8	2.6	13.2
Santander	2585	6.0	5.6	6.0	5.5	12.8
Cundinamarca	1282	3.5	5.8	6.0	1.9	12.8
Risaralda	1161	2.8	7.4	8.9	11.0	21.0
Tolima	763	1.9	4.0	0.0	10.1	21.2
Caldas	710	1.8	6.7	6.8	-0.8	6.6
Bolívar	635	1.7	4.4	2.1	9.8	19.7
Norte de Santander	617	1.4	5.2	6.0	-7.6	2.9
Boyacá	290	0.7	5.8	6.0	1.8	8.8
Quindío	288	0.7	5.2	6.0	6.9	18.1
Nariño	255	0.6	3.4	0.0	14.7	25.1
Huila	182	0.5	4.4	0.0	10.1	19.2
Cauca	155	0.4	3.7	0.0	-0.4	-1.8
Meta	119	0.3	2.8	0.0	-10.5	12.4
Córdoba	126	0.3	3.7	0.3	21.1	25.1
Cesar	72	0.2	1.8	0.0	-11.8	16.9
Magdalena	73	0.2	5.0	1.3	-33.7	-13.2
Sucre	54	0.1	4.5	4.4	-5.7	-11.4
Putumayo	9	0.0	8.4	6.0	22.8	7.1
Caquetá	5	0.0	5.9	6.0	50.7	70.1
Chocó	1	0.0	3.1	3.1	-36.0	-36.0

Prelim Results (Metro Area)

Dependent Variable:	$\mathcal{V}$
Model:	(1)
<i>Variables</i>	
Bogotá D.C., Soacha	0.0485*** (0.0132)
Cali, Yumbo	0.0706*** (0.0174)
Medellín, Valle de Aburrá	0.0440** (0.0141)
Manizales, Villamaria	0.0359 (0.0512)
Barranquilla, Soledad	0.0309 (0.0266)
Bucaramanga, Giron, Floridablanca	0.0654** (0.0226)
Pereira, Santa Rosa de Cabal, Dosquebradas	0.1391*** (0.0355)
Cartagena	0.1402* (0.0758)
Rest of Colombia	0.0244 (0.0197)
<i>Fit statistics</i>	
Observations	32,373
R ²	0.00280
Adjusted R ²	0.00255

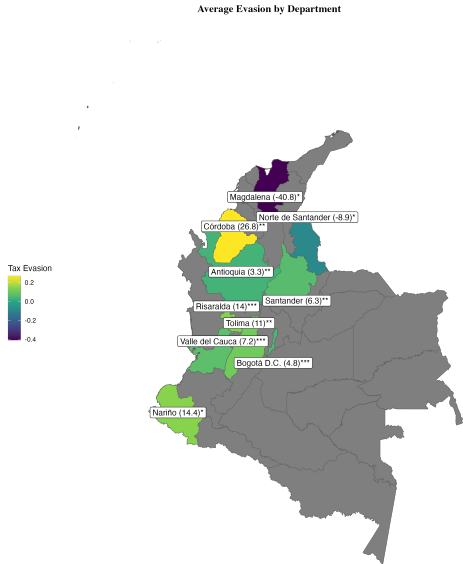
Clustered (plant & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Prelim Results (Departamento)

Dependent Variable: Model:	ψ (1)
<i>Variables</i>	
Antioquia	0.0331** (0.0143)
Atlántico	0.0323 (0.0267)
Bogotá D.C.	0.0484*** (0.0133)
Bolívar	0.1116 (0.0743)
Boyacá	0.0498 (0.0822)
Caldas	-0.0275 (0.0574)
Cauca	0.1291 (0.0779)
Cesar	-0.3661 (0.3888)
Córdoba	0.2684** (0.0882)
Cundinamarca	0.0371 (0.0449)
Huila	0.1464 (0.0816)
Magdalena	-0.4084* (0.2023)
Meta	-0.1142 (0.1742)
Nariño	0.1441* (0.0655)
Norte de Santander	-0.0894* (0.0427)
Quindío	0.0801 (0.0608)
Risaralda	0.1396*** (0.0349)
Santander	0.0629** (0.0227)
Sucre	0.0642 (0.1590)
Tolima	0.1397** (0.0414)
Valle del Cauca	0.0718*** (0.0152)
<i>Fit statistics</i>	
Observations	32,373
R ²	0.00895
Adjusted R ²	0.00834

Clustered (plant & year) standard errors in parentheses
 Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

On the Map



Next Steps

- ▶ Are there any patterns? Do they make sense?
- ▶ Can any of these regions be linked to Free Trade Zones or Special Economic Zones? Since when are these in place?
 - ▶ Zonas Francas: started in 1958. During this period, there were changes made in 1981 and 1985, and 1991. 1981-1984, firms in Zonas Francas were exempt from income and sales taxes, as well as from import duties. They were under strict vigilance by the tax authority and confined to specific locations. Since to be eligible, your products had to be exported, most traditional firms choose not to locate in ZFs. Mostly multinationals and exporting Maquilas took advantage of ZFs. There were ZFs in Barranquilla, Cartagena, Santa Marta, and Buenaventura.
 - ▶ Zonas Económicas Especiales: started in 2020

Appendix

Appendix

B) Value of Reported Ghost Purchases Over Firm Total Purchases, by Firm Size

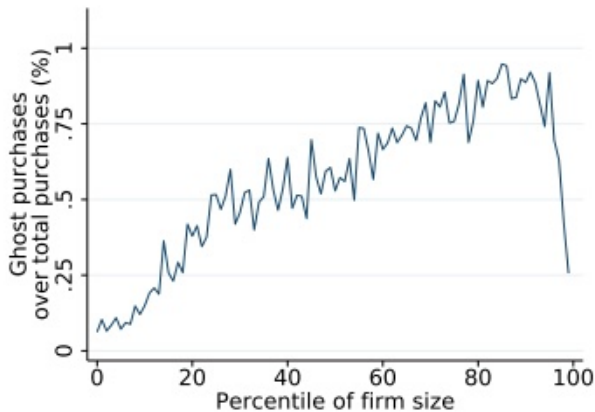


Figure 1: Figure 1B from Carrillo et al. (2022) ↵

Robust Check for Size Heterogeneity

$$\mathcal{V}_{ijt} = \sum_{z \in Z} \sum_{h \in 1,2} \mu_{z,h} x_{z,ijt}^h + \eta_{ijt} \quad (16)$$

Table 1: Coefficient estimates . 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

	IV: Lagged Revenue	OLS: Lagged Revenue
Revenue	1.8012*** [1.5744, 2.0744]	
Revenue ²	-0.7594*** [-1.0265, -0.5149]	
Revenue _{t-1}		1.456*** [1.274, 1.6735]
Revenue _{t-1} ²		-0.5586*** [-0.784, -0.339]
Labour	-0.7149*** [-0.9244, -0.5631]	-0.4281*** [-0.6214, -0.2994]
Labour ²	0.1051 [-0.0561, 0.3035]	-0.09 [-0.2306, 0.1051]
Capital	-0.1124 [-0.277, 0.085]	-0.022 [-0.1776, 0.1737]
Capital ²	-0.2332** [-0.4548, -0.0406]	-0.2997*** [-0.5255, -0.1106]
Exports	-0.1127 [-0.7196, 0.3843]	-0.2759 [-0.8917, 0.192]
Exports ²	0.066 [-0.483, 0.7506]	0.2503 [-0.2569, 0.9386]
Imports	-1.1048*** [-1.4765, -0.7346]	-1.1253*** [-1.5047, -0.7651]
Imports ²	1.3084*** [0.8792, 1.7581]	1.3469*** [0.9235, 1.8055]
Const.	-0.1309*** [-0.1662, -0.0985]	-0.1158*** [-0.1492, -0.0826]

Table 2: Coefficient estimates. Percentile of size of different variables. 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

	IV: Lagged Revenue	OLS: Lagged Revenue
Revenue	1.1491*** [1.0863, 1.2497]	
Revenue _{t-1}		0.9948*** [0.9362, 1.0738]
Labour	-0.6258*** [-0.7001, -0.5599]	-0.5044*** [-0.5691, -0.4402]
Capital	-0.3082*** [-0.3679, -0.2278]	-0.2737*** [-0.3309, -0.1943]
Exports	-0.1004 [-0.5756, 0.283]	-0.2292 [-0.7139, 0.1385]
Imports	-0.7292*** [-0.9883, -0.5182]	-0.7386*** [-1.0034, -0.5334]

Why the Negative Sign for Some Small Firms?

- ▶ Smaller firms are more likely to be Proprietorships and less formal
- ▶ Proprietorships report their industrial activity as part of their individual income taxes. They need to keep track of their personal expenses and their industrial activity expenses
- ▶ Because they are more likely to be less formal than larger firms, they may lack a rigorous expense report system
- ▶ Thus, they might be consistently underreporting their expenses

References

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